Program

```
#include <stdio.h>

#define MAX 50

int main()
{
   int graph[MAX][MAX];
   int visited[MAX];
   int queue[MAX];
```

```
   int n, i, j;
   int front = 0, rear = 0;
   int start, current;

   // Read number of vertices
   printf("Enter number of vertices: ");
   scanf("%d", &n);

   // Read adjacency matrix
   printf("Enter the adjacency matrix:\n");

   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         scanf("%d", &graph[i][j]);
      }
   }

   // Initialize visited array
   for (i = 0; i < n; i++)
   {
      visited[i] = 0;
   }

   // Read starting vertex
   printf("Enter starting vertex: ");
   scanf("%d", &start);

   // Insert starting vertex into queue
   visited[start] = 1;
   queue[rear] = start;
   rear++;

   printf("BFS Traversal: ");

   // BFS traversal
   while (front < rear)
   {
      // Remove vertex from queue
      current = queue[front];
      front++;

      printf("%d ", current);

      // Find adjacent vertices
      for (i = 0; i < n; i++)
```

```
    {
        if (graph[current][i] == 1 && visited[i] == 0)
        {
            visited[i] = 1;
            queue[rear] = i;
            rear++;
        }
    }
  }

  return 0;
}
```

Sample Input

Enter number of vertices: 5

Enter the adjacency matrix:

0 1 1 1 0

1 0 1 0 0

1 1 0 0 1

1 0 0 0 0

0 0 1 0 0

Enter starting vertex: 0

Output

BFS Traversal: 0 1 2 3 4